

Instrumentation and Control

CSE 325

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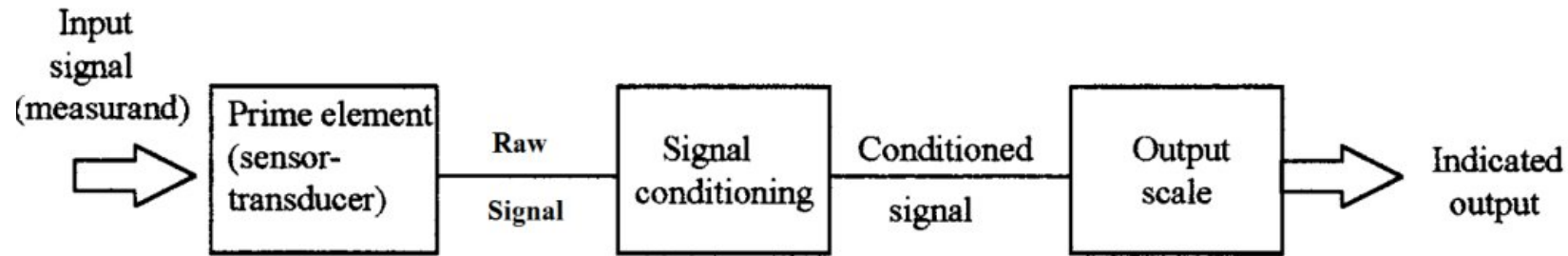
Lecturer(EEE)

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Signal Conditioning Systems



Signal conditioning refers to operations performed on signals to convert to a suitable for interfacing with other elements in the instrumentation process.

- ☁ The raw signal may:
- ☁ be too **small** – have to be **amplified**
- ☁ **contain noise** – have to **be removed**
- ☁ be **non-linear** – need **linearization**
- ☁ be **analog** – have to **make digital**
- ☁ be **digital** – have to **make analog**
- ☁ be **hazardous** – require **isolation**

The conditioned signal is then transferred to the output stage (for indication, recording, logging etc.) which provides the value corresponding to the measurand.

Elements of Signal Conditioning System

The following are some of the processes that can occur in conditioning a signal:

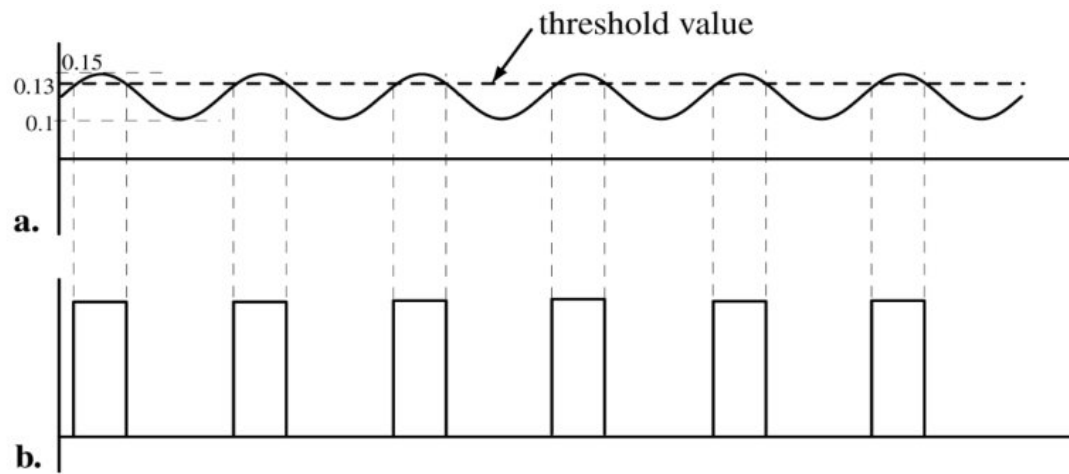
- 🚚 **Conversion** – getting the signal into the right type of signal
- 🚚 **Change of signal level** – getting the level of the signal right
- 🚚 **Eliminating or reducing noise** – increasing the signal-to-noise ratio (SNR)
- 🚚 **Linearization** – making the output linear of the measurand
- 🚚 **Impedance/load matching** – for proper power coupling
- 🚚 **Protection** – to prevent damage to the next element/component

Conversion – getting the signal into the right type of signal

Analog to Digital and Digital to Analog Converters

Threshold Digitization

It is sufficient to use a threshold comparator which will then produce a digital output.

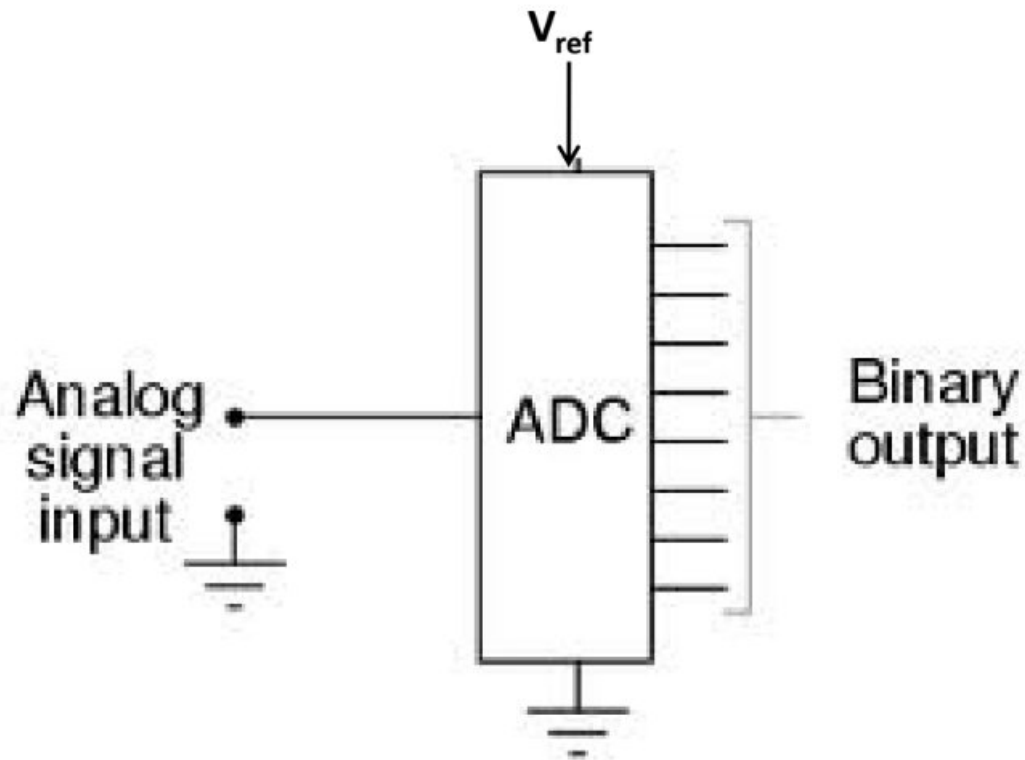


- Output from magnetic pick-up circuit varies from 100 mV to 150 mV (for example).
- The threshold is set by the resistors to 0.13V.
- The output is zero until the voltage on the (+) input rises above the threshold.
- When the input dips below 0.13V the output goes back to zero.

Now, each pulse represents a rotation of the rotor. Counting the pulses in a given time can give the speed of rotation of the rotor or other data (e.g. flow rate)

Analog to Digital Conversion

Converts analog signals into binary numbers/words

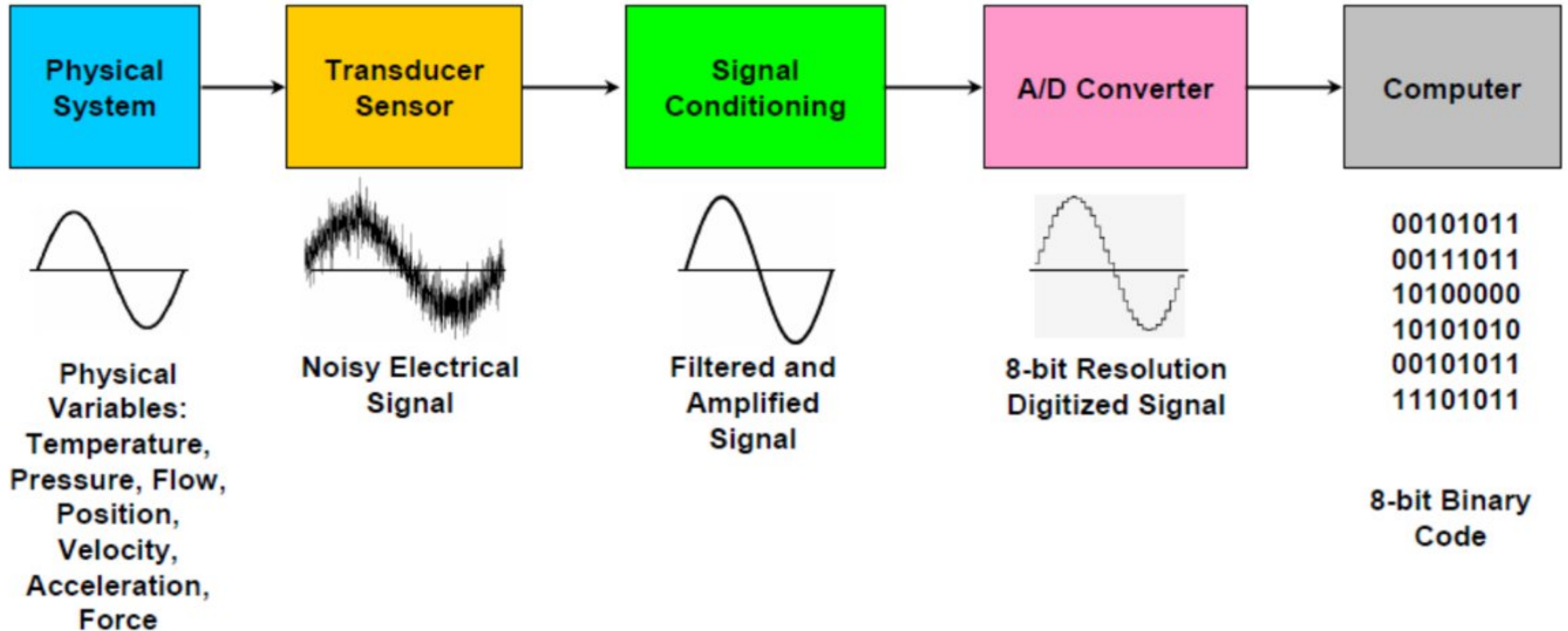


Why the conversion:

- **Processing by a computer** or by logic circuits, including **arithmetic operations, comparison, sorting, ordering, and code conversion**
- **Storage** on electronic media until **ready for further handling**
- **Display in numerical or graphical form** on a computer/other digital device
- **Digital transmission** for **less error**

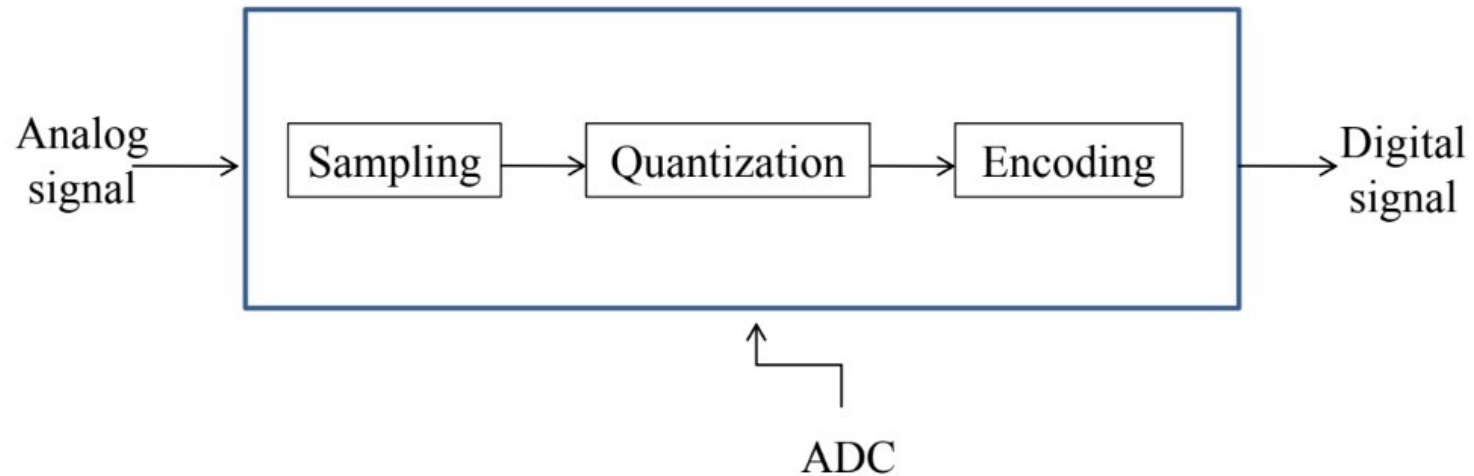
Analog to Digital Conversion

Where do we use an ADC?



Analog to Digital Conversion

Steps in A/D Conversion:



Analog signal: continuous in time and magnitude

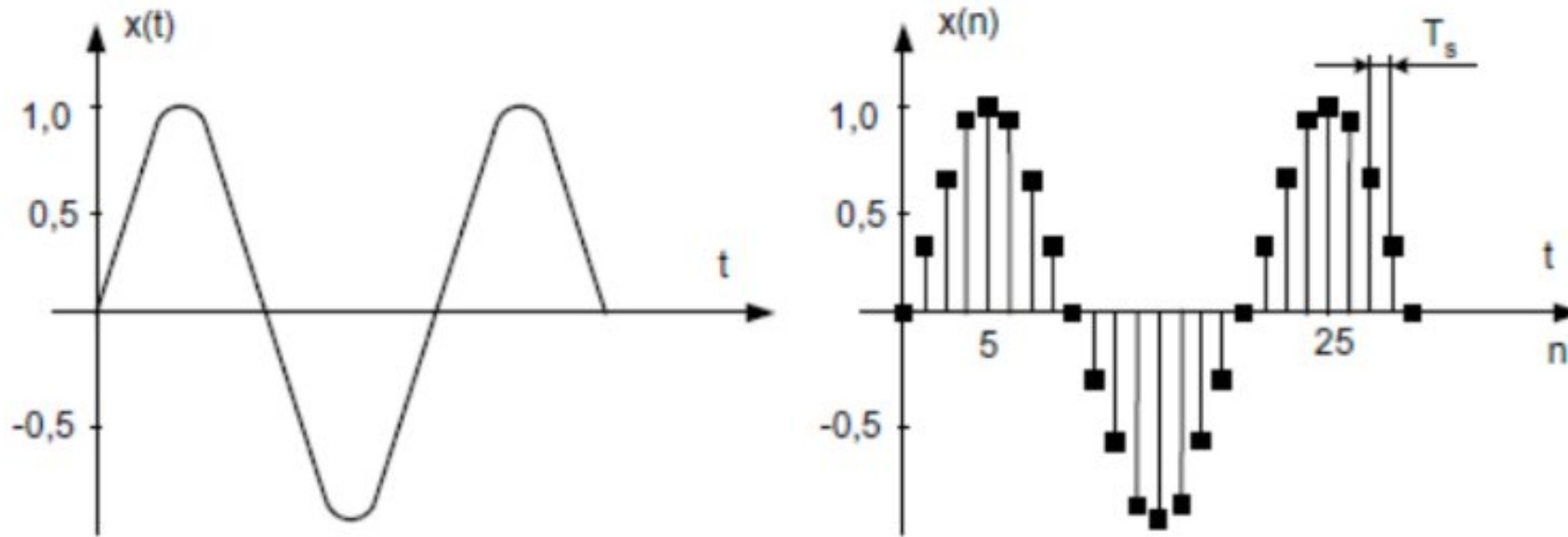
Sampled signal: discrete in time but can be of any magnitude

Quantized signal: discrete in magnitude

Digital signal: coded quantized signal

Analog to Digital Conversion : Sampling

Sampling converts analog signal (continuous in time and magnitude) **into discrete time signal** (discrete in time but can have any magnitude)



The intervals, t_s , must be carefully chosen to ensure an accurate representation of the original analogue signal.

Analog to Digital Conversion : Sampling

The problem of losing information in the sampling process, it is necessary to recall Shannon's information theorem and Nyquist's criteria.

Shannon's information theorem states the following.

- An analogue signal with a bandwidth of f_a must be sampled at a rate of $f_s > 2f_a$

To avoid loss of information.

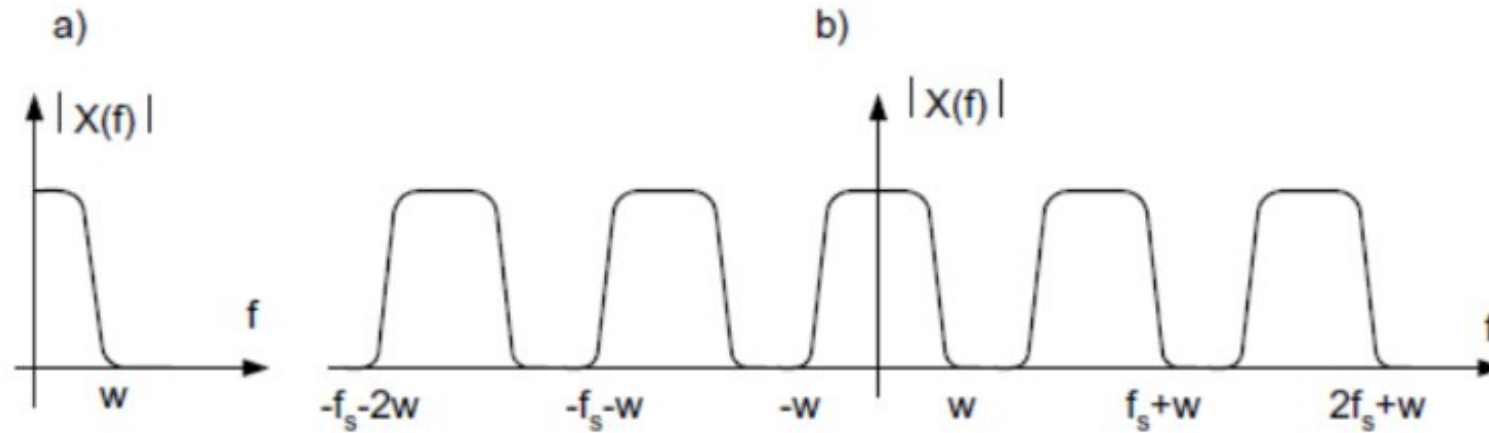
- The signal bandwidth may extend from d.c. to f_a or from f_1 to f_2 , where $f_a = f_2 - f_1$

The Nyquist criterion is as follows.

- If $f_s < 2f_a$, then a phenomenon called aliasing will occur.

To avoid aliasing and loss of information, sampling frequency must be at least twice the highest frequency of the signal.

Analog to Digital Conversion : Sampling

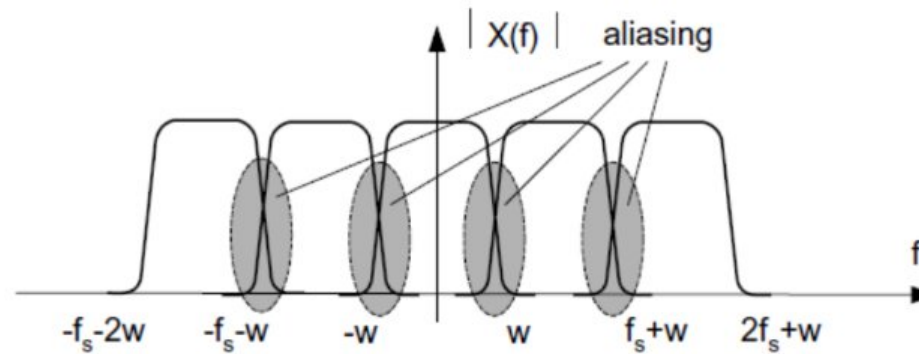


The signals in the above figure were sampled with the frequency $f_s > 2w$.

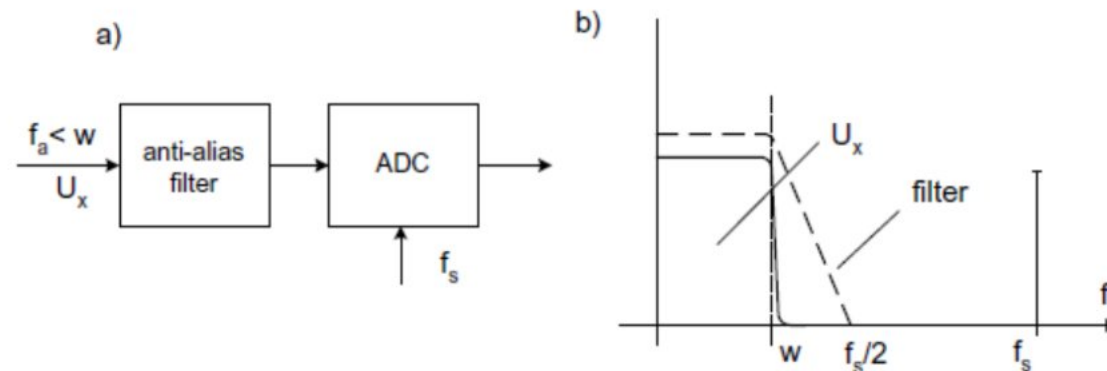
Analog to Digital Conversion : Sampling

Aliasing is the distortion caused when a signal is sampled below the Nyquist rate, resulting in overlap of frequency components.

If $f_s < 2f_a$, the replicated signal bands interfere mutually and a distorted signals appear at the output. This effect is called aliasing.



Anti-aliasing filter: An anti-aliasing low-pass filter is used before sampling to limit the signal bandwidth to the Nyquist frequency ($f_s/2$), thus preventing aliasing.



Analog to Digital Conversion : Quantization

Quantization is breaking down the values of the sampled signal into a set of finite number of quantized values

Example:

We have 0-10V signals. Separate them into a set of discrete states with 1.25V increments. (How did we get 1.25V?)

Quantizer Output States

- The number of output levels (states) in a quantizer is:

$$M = 2^n$$

where:

- n = number of bits in A/D converter

Example

- For a **3-bit ADC**:

$$M = 2^3 = 8$$

So, there are **8 quantization levels**

Quantization Step Size (q)

- It is the **difference between two adjacent levels**

$$q = \frac{V_{\max} - V_{\min}}{M}$$

$$q = \frac{10-0}{8} = 1.25V$$

Output States	Discrete Voltage Ranges (V)
0	0.00-1.25
1	1.25-2.50
2	2.50-3.75
3	3.75-5.00
4	5.00-6.25
5	6.25-7.50
6	7.50-8.75
7	8.75-10.0

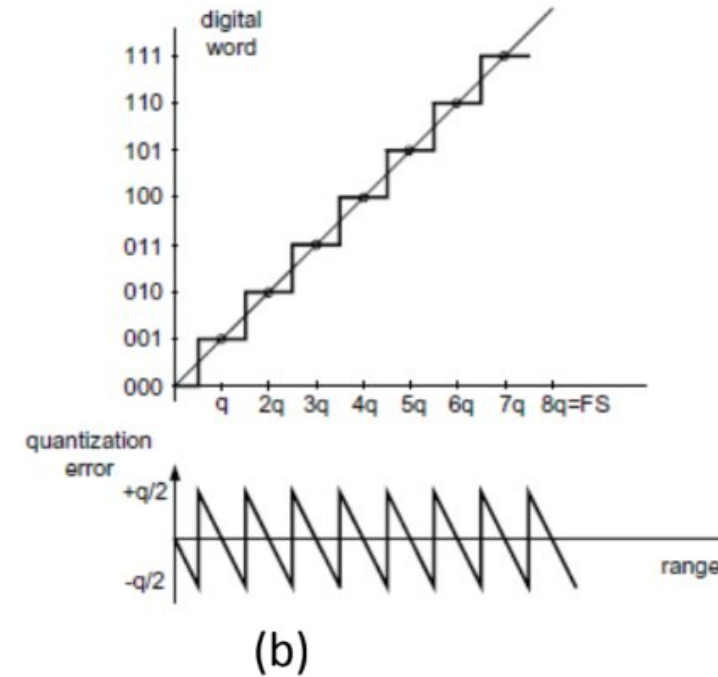
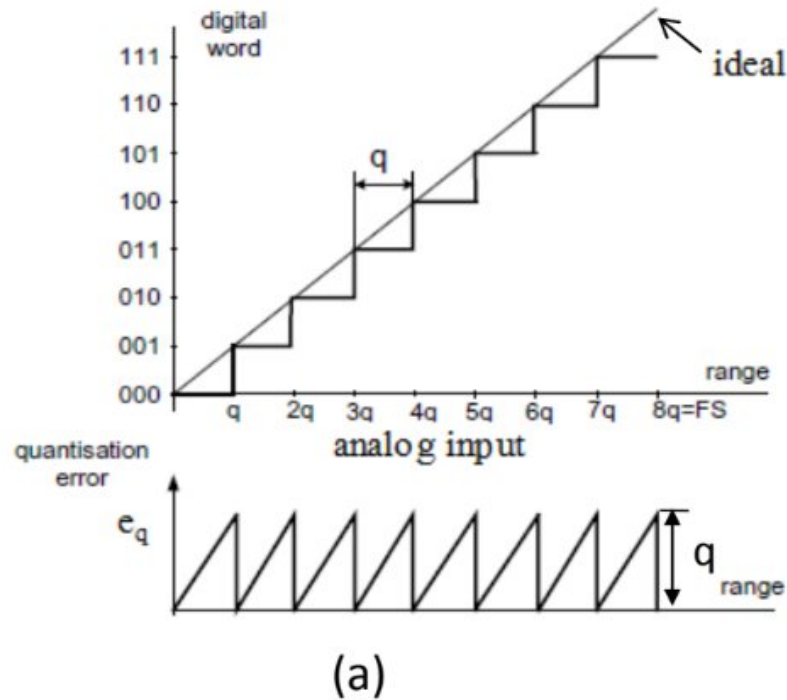
Analog to Digital Conversion : Encoding

Encoding is *the process of converting quantized values into binary (digital) form.*

A unique digital number/word is assigned to each of the quantized states

Output States	Discrete Voltage Ranges (V)	Binary code
0	0.00-1.25	000
1	1.25-2.50	001
2	2.50-3.75	010
3	3.75-5.00	011
4	5.00-6.25	100
5	6.25-7.50	101
6	7.50-8.75	110
7	8.75-10.0	111

Analog to Digital Conversion : Transfer characteristics



From Fig (a) we can see that the quantization error is **varying between 0 and q Value**.

Usually the characteristic of the quantization is shifted by the $q/2$ value thus **the error of quantization is varying between $-q/2$ and $+q/2$** (Fig. b).

Analog to Digital Conversion : Transfer characteristics

Voltage Ranges	Quantized state	Binary code	Analog equivalent	Error range
$0 - q$	0	000	0.00	$0 - q$
$q - 2q$	1	001	q	$0 - q$
$2q - 3q$	2	010	$2q$	$0 - q$
$3q - 4q$	3	011	$3q$	$0 - q$
$4q - 5q$	4	100	$4q$	$0 - q$
$5q - 6q$	5	101	$5q$	$0 - q$
$6q - 7q$	6	110	$6q$	$0 - q$
$7q - 8q$	7	111	$7q$	$0 - q$

Voltage Ranges	Quantized state	Binary code	Analog equivalent	Error range
$0 - 0.5q$	0	000	0.00	$0 - q/2$
$0.5q - 1.5q$	1	001	q	$(-q/2) - (+q/2)$
$1.5q - 2.5q$	2	010	$2q$	$(-q/2) - (+q/2)$
$2.5q - 3.5q$	3	011	$3q$	$(-q/2) - (+q/2)$
$3.5q - 4.5q$	4	100	$4q$	$(-q/2) - (+q/2)$
$4.5q - 5.5q$	5	101	$5q$	$(-q/2) - (+q/2)$
$5.5q - 6.5q$	6	110	$6q$	$(-q/2) - (+q/2)$
$6.5q - 7.5q$	7	111	$7q$	$(-q/2) - (+q/2)$